

ZN Infotech Pvt. Ltd.

Cloud Computing Syllabus

Month 1: Cloud Fundamentals & IT Foundations

Introduction to cloud computing, evolution of IT infrastructure, cloud characteristics, service models (IaaS, PaaS, SaaS), deployment models, virtualization concepts, networking basics including IP addressing, DNS, CIDR, subnets, and cloud architecture principles.

Month 2: Core Cloud Services & Architecture

Compute services (virtual machines, instance types, auto scaling, load balancing), storage services (object, block, file storage), database services (SQL and NoSQL), backup and restore mechanisms, high availability, and fault-tolerant cloud design.

Month 3: Linux & DevOps Foundations

Linux operating system fundamentals, file system, permissions, users and processes, shell scripting basics, DevOps principles, software development lifecycle, version control using Git and GitHub, and CI/CD concepts.

Month 4: Containers, CI/CD & Orchestration

Containerization using Docker, Dockerfile and image management, container registries, continuous integration and deployment pipelines, automation tools, Kubernetes fundamentals including pods, deployments, services, and cloud-native application delivery.

Month 5: Cloud Security, Monitoring

Introduction to cloud computing, evolution of IT infrastructure, cloud characteristics, service models (IaaS, PaaS, SaaS), deployment models, virtualization concepts, networking basics including IP addressing, DNS, CIDR, subnets, and cloud architecture principles.

Month 6: Advanced Cloud Concepts & Real Projects

Advanced networking concepts, virtual private cloud design, serverless computing, event-driven architecture, cost optimization, disaster recovery strategies, real-world cloud and DevOps projects, deployment best practices, and career-oriented outcomes.